

This is the Sedimentologica L^AT_EX template example and build test

3 Abstract

This is a test abstract used for evaluating the build of an article using the *Sedimentologica* L^AT_EX template using LuaL^AT_EX. It contains no meaningful scientific content, but contains useful guidelines for writing your article using the template.

If your author and affiliation lists and CRediT roles are long, the abstract may run over the first page in 'preprint' mode. If this is the case, please attempt to shorten your abstract; however, if it is not reasonably possible please contact support@sedimentologica.org. Use `\documentclass[preprint]{sedimentologica}` to test the length of your abstract.

11 Plain language summary

This file tests the *Sedimentologica* article template build using LuaL^AT_EX. A plain language summary is mandatory.

14 Translations

We encourage authors to provide a translation of your abstract and plain language summary into a language of your choice. If you wish to provide multiple translations, please contact the editorial team for guidance

If your translation language requires special fonts (e.g., Arabic), instructions are available in the template file and the README.md file. Should you need further help with this, please contact the editorial team support@sedimentologica.org.

20 1 Introduction

As of 2026, *Sedimentologica* has moved to L^AT_EX to typeset accepted publications in the journal. The primary motivations for this switch are to:

- use an open-source, free software for typesetting,
- provide authors with a streamlined pre-print to publication workflow,
- lower the time investment required by our volunteer typesetting team, and,
- ensure all papers published in *Sedimentologica* have a unified design.

While the primary uses of the template are for typesetting by the *Sedimentologica* team and authors preparing their article for submission to *Sedimentologica* it also provides you with an easy way to generate a pre-print version that you can post on a pre-print server (e.g., [EarthArXiv](#)). Simply use `\documentclass[preprint]{sedimentologica}` in the preamble above.

Like the policy of *Seismica*, we do not require authors to use this template at initial submission. **However**, **minimum formatting requirements** apply and upon acceptance authors using L^AT_EX should submit their final version using the *Sedimentologica* L^AT_EX template. Not using the template may incur delays in copy-editing, proofing, and publication, For authors using Microsoft Word or LibreOffice etc, there may be greater delays incurred depending on the complexity of the document.

2 Minimum formatting and layout requirement

While *Sedimentologica* strongly encourages the use of our templates for articles submitted to the journal, you may submit your article without using the templates if the following conditions are met:

- Your article is written in a single dialect of English
- Your manuscript is a single PDF, DOC, DOCX, or ODT file containing all figures and in-text tables.
- Your data is made accessible via a repository such as Zenodo.
- You have added to your manuscript the following statements, after the conclusions and before the references
- Acknowledgements
- Data and code availability
- Conflict of interest statement
- Artificial intelligence use statement
- Funding declaration

3 Language

All submitted documents must be in English. Authors are required to use a single variety of English consistently throughout the manuscript, such as Australian, British, Canadian, Commonwealth, or US English.

While *Sedimentologica* requires manuscripts to be written in English, we encourage authors to provide an abstract and plain language summary translation in their preferred language of choice.

Submitted documents should be thoroughly proof-read prior to submission to ensure they are free from grammatical and spelling errors.

Abbreviations that are not standard to the field are not to be used in the abstract and should be introduced in the corpus of the text. The use of standard abbreviations such as “Fm.” or “Mbr” respectively, for “Formation” or “Member” should be reserved for figures and tables, while full-length words are used in the corpus of the manuscript.

All units must be from the [International System of Units](#) or derivative thereof (i.e. metres rather than feet or kPa rather than PSI). Customary/non-SI units may be included in parentheses.

Utilise the commands and options available in [siunitx](#) for any quantity or number.

- `\ang[options]{angle}`
- `\num[options]{number}`
- `\unit[options]{unit}`
- `\qty[options]{number}{unit}`
- `\numlist[options]{numbers}`
- `\numproduct[options]{numbers}`
- `\numrange[options]{numbers}{number2}`
- `\qtylist[options]{numbers}{unit}`
- `\qtyproduct[options]{numbers}{unit}`
- `\qtyrange[options]{number1}{number2}{unit}`

3.1 Inclusive Language

[Sedimentologica](#) strongly encourages authors and co-authors to use inclusive and non-discriminatory language in their manuscripts. The American Psychological Association (APA) has produced a guide you may find useful for this purpose, which you can obtain from [apa.org](#).

Scientific research is always a team effort. Therefore, we encourage using “we” about the authors as much as using passive voice or third person.

78 *Sedimentologika* encourages authors to use both traditional place names and their English equivalents, where
79 appropriate, to acknowledge the cultural significance and diversity of the regions being described, e.g., "the
80 Kaikōura Canyon is located east of Te Wāhipounamu (the South Island of Aotearoa/New Zealand)".

81 When applicable, acknowledgement of traditional custodians and their lands should be added.

82 3.2 Politically Sensitive Geographic Names

83 As Earth Scientists, we often use geospatial data. In some cases the geographic name of a location may be
84 disputed, or the border of a territory has not been honoured or is disputed.

85 For cases of geographic name disputes (e.g. Gulf of Mexico/Gulf of America), authors should first determine if an
86 arbitration decision has been made, for example under the UNCLOS, and use that name. If no such decision has
87 been made, authors should use the name which is most recognised internationally, refer to appropriate United
88 Nations maps (e.g. [UN Map of the World](#)), or provide both names. The intent is to ensure the name used is as
89 neutral as possible.

90 3.3 Geologic Time

91 For geologic time, please use nomenclature of the International Chronostratigraphic Chart, the latest version can
92 be found at [text](#).

93 4 Figures and tables

94 This section outlines the general guidelines for figures and tables. Please refer to *Sedimentologika's* [figure and](#)
95 [table guidelines](#) for further detail.

96 The final document will have a two-column page layout. You can test the appearance of your tables and figures in
97 this layout by using the preprint option for the sedimentologika document class.

98 4.1 Dimensions

99 It is best practice to design and test your figures and tables at their intended final size. Figures and tables should
100 preferably be designed to fit within either a single column (e.g., [Figure 1](#)) or across two columns (e.g., [Figure 2](#))
101 in portrait page mode. However, landscape figures and tables are allowable, but only use these if absolutely
102 required for legibility. [Table 1](#) summarises the maximum dimensions allowable for figures and tables that fit
103 within a single page.

Intended Width	Maximum Width	Maximum Height
one column	86 mm	230 mm
two column	180 mm	230 mm
landscape	257 mm	160 mm

Table 1 – Summary of standard dimensions for figures and tables.

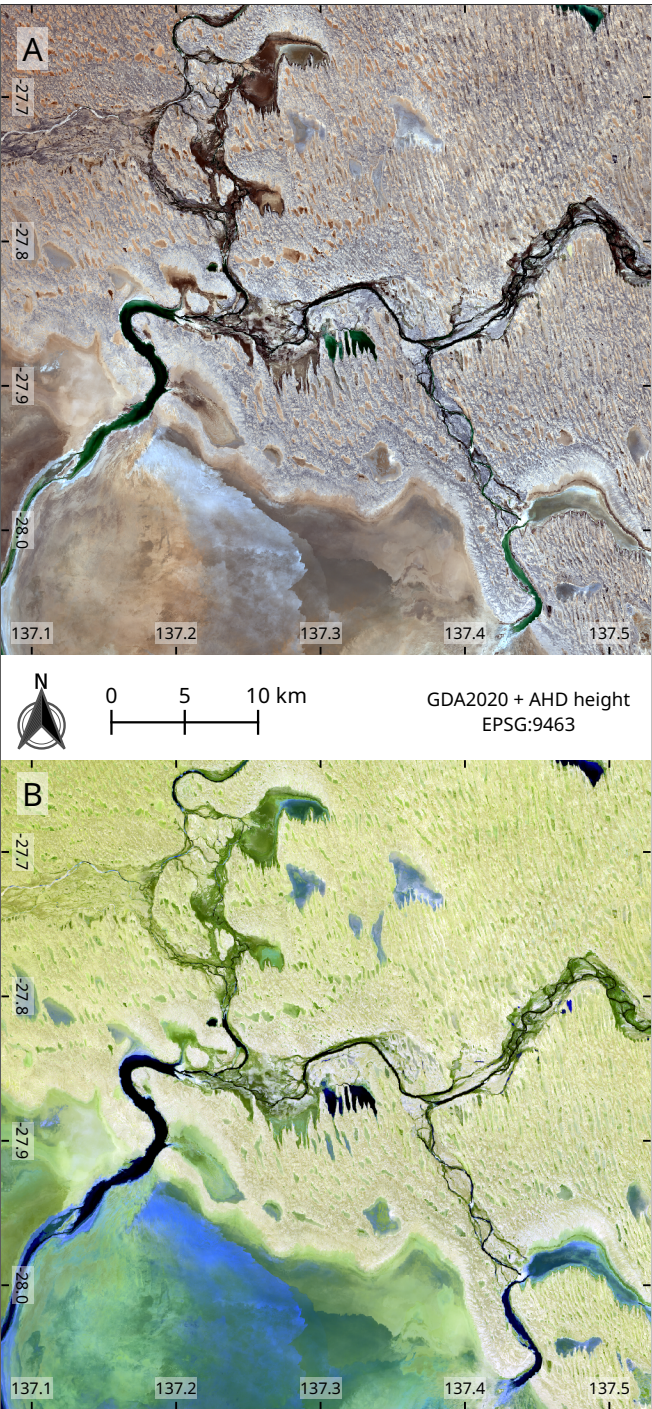


Figure 1 – Sentinel 2 L2A imagery (2025-10-18) of north-eastern Kati Thanda-Lake Eyre at the Warburton River discharge. **(A)** True colour image mapping bands 4, 3, and 2 to R, G, and B channels. **(B)** False colour image mapping bands 12, 11, and 2 to R, G, and B channels. Contains modified Copernicus Sentinel data [2025].

104 If a figure is being placed at the end of the document rather than near its insertion point define the figure optional
105 position argument using a ! or use \clearpage directly after the figure environment.

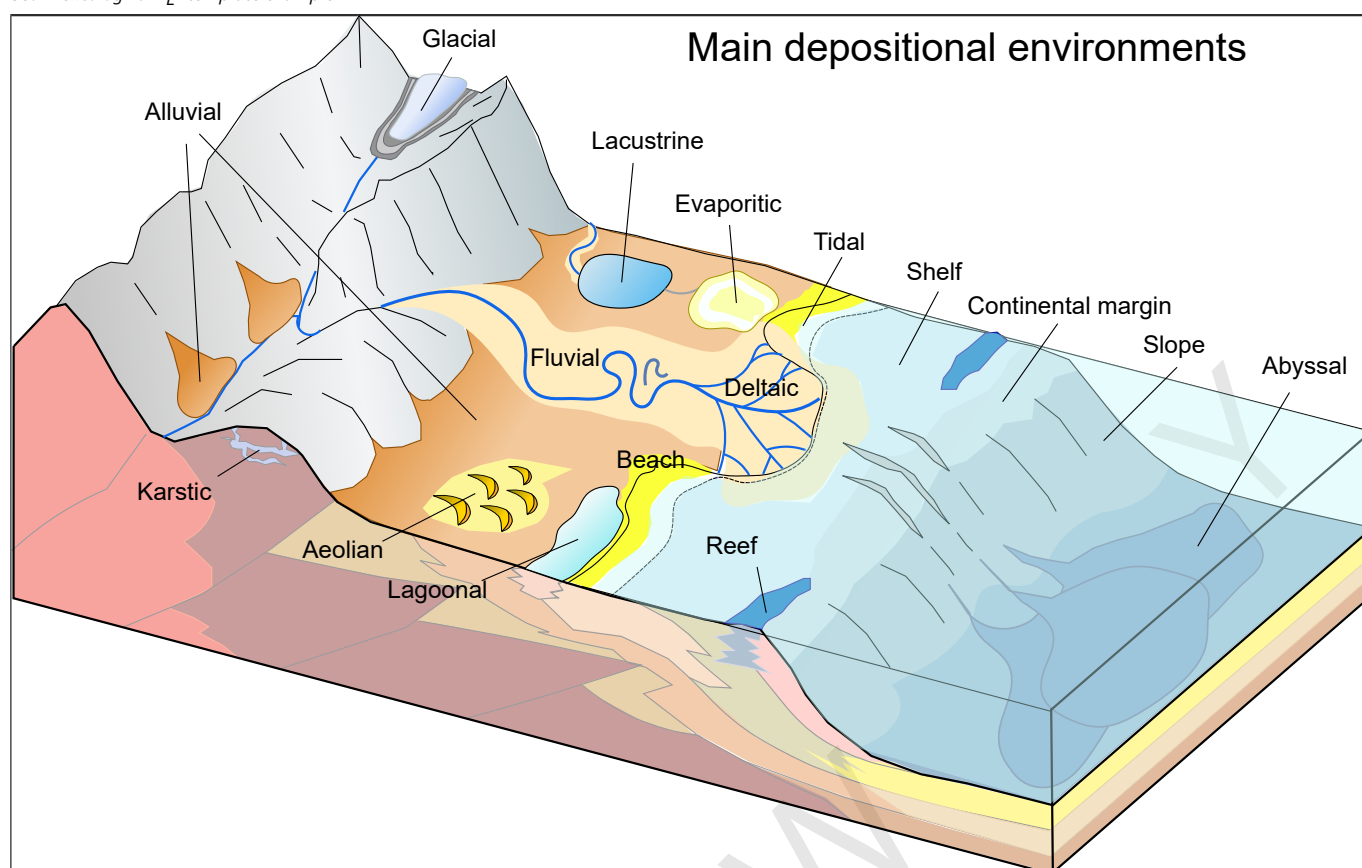


Figure 2 – Schematic diagram of the main depositional environments in sedimentary systems. Source: Mikenorton (2018)

4.2 Figure design

Figures must be designed with accessibility in mind. Do not rely on colour alone, use symbols or patterns to assist with differentiation on graphs and other diagrams.

Preferably use the [Noto Sans](#) font family as that is what we use for final production. However, any standard, sans-serif font (e.g., Arial, Helvetica, Verdana) can be used. Be consistent and only use one font family.

Use at most three (preferably no more than two) different font sizes for annotations on figures. The standard font size should be 10 pt Noto Sans or equivalent. Regardless of font choice, the minimum character height must be no less than 2.8 mm at final production size for accessibility purposes. Aim to use larger font sizes (e.g., 10 pt Noto Sans) to improve accessibility.

Colour schemes **must** be colour-vision deficiency safe (e.g. [Scientific Colour Maps](#)).

4.3 Maps

Map figures must have a scale, grid markers of some form, and state the coordinate reference system of their projection. [Figure 1](#) is an exemplar of minimum requirements for map figures.

Due to the nature of geology being an international science, some maps may have areas of disputed borders or regions. In these cases, please use the internationally recognised border or region where possible.

4.4 Indexing and cross-referencing

Figures and tables should be numbered in order of appearance and cross-referenced in text. In L^AT_EX, use `\autoref{ref}` to handle this. For example, these are cross-references to [Figure 2](#) and [Table 1](#) using `\autoref`.

For figures with sub-panels, add uppercase letters for indexing purposes as shown in [Figure 1A](#) and [Figure 1B](#). If additional sub-indices are required, use parenthesised lowercase Roman numerals, e.g., A(i).

4.5 Table formatting

Complex tables present a significant challenge in L^AT_EX for our volunteer production team. Please only use tables where needed, and keep them simple. Complex tables will incur production delays.

Use a table only if there isn't a simpler way to present your content, such as a paragraph of text or figure. Each table must be independently understandable, with a clear title or caption, defined abbreviations, and sufficient explanatory notes where needed. Tables should be numbered sequentially in the order they are cited in the manuscript. Design tables for readability and accessibility. The minimum font size should be Noto Sans 8 pt (or equivalent). Keep tables simple:

- use simple structure, clear column headings, consistent terminology, and SI units;
- avoid excessive decimal places, duplicated information, empty cells without explanation, and unnecessary formatting;
- do not use colour alone to communicate meaning.

Large or highly detailed datasets should be deposited in an appropriate repository and cited in the data availability statement rather than included as oversized manuscript tables. Datasets provided in a form such as CSV, ODS, XLSX, RData, JLD2, HDF5 etc. are better for study reproducibility.

Tables rules are formatted using [booktabs](#). This means you can use the `\toprule`, `\midrule`, and `\bottomrule` commands within the `tabular` environment.

The [nicematrix](#) package is also loaded, providing customisation for blocks and the X for the flexible column widths.

145 **Table 1** is a simple example of a one-column wide table with correct styling and using the NiceTabular*
 146 environment.

147 To ensure correct widths of tables, please use the provided custom commands

148 · \onecolwidth for single-column width

149 · \twocolwidth for two-column width

150 as the size of your NiceTabular* or tabular* environment.

151 For tables that will extend beyond a single page, longtable is available. If the longtable needs to be two
 152 columns wide, or in landscape orientation, please see the example_longtable.tex for an example of how to
 153 format these. We have placed the longtable at the end of this build to not clutter the primary information.

154 To place a table that is only a single column wide (i.e. small tables), use the table environment.

155 Otherwise, use the table* environment to place a table that will span both page columns.

156 *NOTE:* nicematrix and longtable will not work in the same table.

157 5 Mathematics

158 In general follow the [Short Math Guide \(PDF\)](#).

159 Use \vec{x} for vectors and \mathbf{M} for matrices, e.g.

160 \vec{y} = \mathbf{X}\vec{\beta} + \epsilon

161 will print:

$$\vec{y} = \mathbf{X}\vec{\beta} + \epsilon \quad (1)$$

162 In-line equations will look like this: $x^2 + y^2 = z^2$, while basic display equations will look like the following:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{i=n} x_j = \frac{x_1 + x_2 \dots + x_n}{n} \quad (2)$$

163 Multiline equations will look as follows:

$$\hat{\rho}_{\partial\rho}^2(\rho^2) = 1 - \frac{n-3}{n-(k+1)-1}.$$

$$(1-\rho^2) {}_2F_1\left(1, 1; \frac{n-(k+1)-1}{2}; 1-\rho^2\right) \quad (3)$$

164 Finally, aligned equations will look like this:

$$\sum (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (4)$$

$$\sum x_i (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (5)$$

$$\sum x_i^2 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (6)$$

$$\sum x_i^3 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (7)$$

165 6 Chemical equations

166 The `mhchem` package is enabled in the template. Use `\ce{}` to wrap your chemistry. For example, `\ce{^{87}_{37}Rb+}`
 167 prints $^{87}_{37}\text{Rb}^+$.

168 7 Referencing

169 Appropriate citation of work is mandatory. While peer-reviewed literature should be the primary sources on which
 170 you rely, material considered to be *grey literature* such as pre-prints, theses, government reports, conference
 171 papers etc. should be appropriately cited.

172 The *Sedimentologica* class uses the `citation-style-languauge` package for referencing.

173 Use `\cite{}` for parenthetical citations, e.g., (Strata, 2065), and `\textcite{}` for in-line text citations, e.g., Helios
 174 et al. (2072) stated that this is an awesome template. Prefixes, suffices, and pointers can be added as optional
 175 arguments, e.g., `\cite[prefix = {e.g., }, suffix={and references therein}] {ExampleThesis}`
 176 will produce (e.g., Diamond, 2060 and references therein).

177 The following is to enable generation of a reference list with examples for an article (Helios et al., 2072), book (Strata,
 178 2065), book section (Core, 2068), dataset (Cryo & Ice, 2075), map (Carto, 2069), preprint (Plume & Geyser, 2074),
 179 report (Agency for Outer Worlds Geology, 2070), software (OrbitLab, 2073) and thesis (Diamond, 2060).

180 8 Conclusions

181 This is a mandatory section, replace this text.

182 Thanks for using the *Sedimentologica* $\LaTeX$ template for preparing your article.

183 Acknowledgements

184 Replace this text with your acknowledgements.

185 Data and code availability

186 Data used in this paper is hosted on Zenodo at doi.org/10.1000/182, \cite{ExampleCitation}.

187 The code used in this paper is available via [and](#) archived on Zenodo at doi.org/10.1000/182, \cite{ExampleCitation}.

188

189 Supplementary materials

190 If you have any additional supplementary material hosted in a repository for this manuscript please replace this with a state-

191 ment linking to the doi, and add the citation, e.g. Supplementary material for this paper can be found at doi.org/10.1000/182

192 \cite{ExampleCitation}.

193 Conflict of interest statement

194 The authors declare they do not have any tangible or perceived conflicts of interest with regard to this research.

195 Artificial intelligence use statement

196 All research, analysis, writing, and editing were performed without generative AI assistance.

197 Funding

198 Declare sources of funding for this research.

199 References

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Table 2 – Summary of laser conditions and materials analysed in each session.

Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2020-02-24	zircon	29	5	2		30
2020-02-24	glass	29	5	3.5		30
2020-02-24	glass	51	5	3.5		30
2020-02-26	zircon	29	5	2		30
2020-02-26	glass	29	5	3.5		30
2020-02-26	glass	51	5	3.5		30
2020-05-06	zircon	19	5	2		40
2020-05-06	glass	19	5	3.5		40
2020-05-06	glass	51	5	3.5		40
2020-05-08	glass	43	5	3.5		40
2020-05-11	zircon	29	5	2		40
2020-05-11	glass	29	5	3.5		40
2020-05-11	glass	51	5	3.5		40
2021-03-30	zircon	29	5	2		30
2021-03-30	glass	43	5	3.5		30
2021-03-31	zircon	30	5	2		40
2021-03-31	glass	43	5	3.5		40
2021-05-06	zircon	30	5	2		40
2021-05-06	glass	43	5	3.5		40
2021-09-06	zircon	30	5	2		30
2021-09-06	glass	43	5	3.5		30
2021-09-06	monazite	20	5	2		30
2022-01-19	baddeleyite	20	5	2		30
2022-01-19	zircon	20	5	2		30
2022-01-19	glass	43	5	3.5		30
2022-01-19	apatite	30	5	3.5		30
2022-02-01	monazite	13	5	2	1.9	30
2022-02-01	xenotime	13	5	2	1.9	30

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Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2022-02-01	glass	43	5	3.5	3.6	30
2022-04-01	glass	43	5	3.5	3.4	30
2022-04-21	apatite	30	5	3.5	3.4	30
2022-04-21	zircon	30	5	2	2	30
2022-04-21	glass	43	5	3.5	3.4	30
2022-05-31	glass	43	5	5	5.2	30
2022-05-31	monazite	13	5	2	1.9	30
2022-05-31	xenotime	13	5	2	1.9	30
2022-06-20	zircon	30	5	2	2.1	30
2022-06-20	glass	43	5	3.5	3.6	30
2022-06-20	xenotime	13	5	2	2.1	30
2022-06-29	glass	43	5	3.5	3.4	30
2022-06-29	xenotime	13	5	2	1.9	30
2022-07-08	glass	43	5	3.5	3.6	40
2022-07-08	rutile	43	5	5	5.2	40
2022-08-30	monazite	20	5	2	2	30
2022-08-30	zircon	20	5	2	2	30
2022-10-10	glass	43	5	3.5	3.4	30
2022-10-10	zircon	30	5	2	1.9	30
2022-10-11	glass	30	5	3.5	3.4	30
2022-10-11	zircon	43	5	2	1.9	30
2022-12-09	glass	43	5	3.5	3.5	40
2022-12-09	rutile	43	5	5	4.9	40
2023-02-20	apatite	30	5	3.5	3.5	30
2023-02-20	glass	30	5	3.5	3.5	30
2023-02-20	zircon	30	5	2	1.9	30
2023-03-22	monazite	13	5	2	1.8	30
2023-03-22	xenotime	13	5	2	1.8	30

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Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2023-04-21	glass	43	5	3.5	3.5	30
2023-04-21	apatite	43	5	3.5	3.5	30
2023-04-21	zircon	30	5	2	2	30
2023-05-09	glass	43	5	3.5	3.5	40
2023-05-23	glass	43	5	3.5	3.6	40
2023-05-29	glass	43	5	3.5	3.4	30
2023-05-29	monazite	13	5	2	2.1	30
2023-06-30	glass	30	5	3.5	3.4	30
2023-06-30	apatite	30	5	3.5	3.4	30
2023-07-11	glass	43	5	3.5	3.5	40
2023-07-11	zircon	20	5	2	1.9	40
2023-11-24	apatite	30	5	3.5	3.5	30
2023-11-24	baddeleyite	30	5	2	2	30
2023-11-24	monazite	20	5	2	2	30
2023-11-24	rutile	43	5	5	4.9	30
2023-11-24	titanite	43	5	5	4.9	30
2023-11-24	xenotime	20	5	2	2	30
2023-11-24	zircon	30	5	2	2	30
2023-11-24	glass	30	5	3.5	3.5	30
2024-04-29	apatite	30	5	3.5	3.5	40
2024-04-29	baddeleyite	30	5	2	2.1	40
2024-04-29	cassiterite	43	5	5	5.1	40
2024-04-29	monazite	20	5	2	2.1	40
2024-04-29	rutile	43	5	5	5.1	40
2024-04-29	titanite	43	5	5	5.1	40
2024-04-29	xenotime	20	5	2	2.1	40
2024-04-29	zircon	30	5	2	2.1	40
2024-04-29	glass	30	5	3.5	3.5	40